

Compact Generative Models of Human Interception Movements

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Abstract

Human interception requires coordinating movement with a moving target, yet repeated attempts vary within and between people. We investigate whether a compact participant representation can support a model of this variability. Using 4,732 free-eye-movement trials from 28 participants, we compared variational autoencoders (VAE), conditional VAE (CVAE), conditional autoencoders (CAE), and spline coefficients compressed by principal component analysis (PCA), with three or eight latent coordinates. Four participant-held-out rotations evaluated reconstruction and generation from separate context trials. Spline + PCA achieved the lowest mean reconstruction error. For generation at eight dimensions, VAE and CVAE improved on spline + PCA in both kinematic-feature and full-trajectory distribution comparisons. Their full-trajectory energy discrepancies were 0.0548 and 0.0549, compared with 0.0981 for spline + PCA; their difference was not significant. These advantages over spline persisted after a training-only redundancy screen reduced eleven features to nine. Replacing a participant's own latent center worsened feature fidelity, although resampling stored context trajectories achieved lower mean path discrepancies than the compact models. An interactive dashboard exposes the evaluated models, with VAE at eight dimensions as the practical default. The results support useful session-specific compression while identifying movement information that the compact representation does not fully recover.

1 Introduction

Repeated interception movements differ in their initiation, duration, curvature and speed profile. A model that predicts only an average trajectory omits this variability. Our objective is to model distributions of detailed interception movements and their kinematic features using a compact representation of an individual participant. Such a representation could support interactive exploration of generated movements while reducing the amount of participant-specific information that must be retained.

We define a participant fingerprint as the average latent code of that person's context trials. Three coordinates address the original compact-representation objective, while eight provide a higher-capacity comparison. The fingerprint is an operational summary of recordings, without assuming that individual latent coordinates correspond to named movement strategies. A useful representation should retain observed paths and support generation of plausible variation on separate trials from a participant who did not contribute to model training.

Variational autoencoders learn nonlinear representations under a probabilistic objective [1], and conditional variants also receive task variables [2]. We compare these approaches with a deterministic neural autoencoder and a statistical pipeline that fits smooth spline coefficients before PCA. The comparison asks whether the representation and its associated generation procedure offer advantages over a simpler statistical model.

The study follows three questions. First, how accurately can each model reconstruct an observed trajectory after compression? Second, can context-derived samples reproduce both kinematic-feature distributions and complete trajectory distributions? Third, does the correct personal center improve generation? These experiments serve different purposes: reconstruction evaluates information retained from an observed trial, generation evaluates variability without receiving the query path, and center replacement tests the contribution of personal context.

The deliverable is a dashboard for comparing and exploring the evaluated models. Its default follows the generative task, rather than the reconstruction ranking alone. The main report presents the modeling ideas, visual examples and principal comparisons. The appendix provides full equations, matrix dimensions, training details, statistical estimators and supporting analyses.

2 Experimental data and trajectory representation

Jason Friedman's laboratory supplied finger-tracking recordings from an interception task. Participants moved from a starting region toward a fixed interception region as a circular target approached horizontally (Figure 1). The target appeared at one of three starting categories on either side. Each category had an associated speed range, so starting position and speed were related by the experimental design. We analyzed the free-eye-movement condition. Model conditions encoded starting category with three one-hot values, starting side with one binary value and executed speed with one continuous value.



Figure 1. Left: task illustration from the supplied display, with a lower starting region and upper interception region. Right: all 180 retained trials from subject01, selected as the first participant in identifier order. Trajectories are filtered and translated to a common start; the x-y plot uses equal spatial scale. These are recordings, not model outputs.

The available dataset contained 4,763 trials from 28 participants. We excluded four trials without an arrival event and 27 with arrival more than one second after the target window ended, leaving 4,732 trials. Twenty-six participants contributed 171-180 retained trials each; the remaining two contributed 60 and 28 from 60 and 29 available recordings. The median retained count was 179.5. Early arrivals and fixation-warning labels remained in the analysis.

Finger position was recorded in three dimensions at 240 Hz. After regularizing the sampling grid and applying a fourth-order 10 Hz forward-backward low-pass filter, we selected the interval from target-motion onset to the end of the recorded trajectory. This window includes waiting and any movement already underway when the target starts moving. Movement onset was identified by a sustained three-frame crossing of 5 tracker units/s in three-dimensional speed. Zero-initiation trials remained included.

Different recording lengths required a common input representation. A cubic interpolant mapped each filtered interval to 100 equally spaced phases, $u = j/99$ for $j = 0, \dots, 99$, before subtracting its first position. Phase therefore describes relative elapsed time rather than distance traveled. The planar x-y coordinates formed a 100×2 array, flattened into 200 values for neural models. This operation translates trajectories without rotating or scaling their spatial amplitude. Physical initiation and movement durations were retained as separate prediction targets.

Four fixed participant rotations used 17 training, four validation and seven test participants each. Every participant appeared in a test set once. Training data determined learned transformations and model parameters; validation data selected settings and checkpoints. Test participants supplied context recordings for enrollment but did not contribute to population-model fitting. Appendix A1 describes the preprocessing and inclusion details.

3 Models, generation and evaluation

3.1 The compression pipelines

Spline + PCA first fitted cubic B-splines with five interior knots. Nine basis coefficients described each coordinate, giving 18 coefficients per trial. Stacking training trials produced an $N \times 18$ coefficient matrix (N training trials), with one trial per row. PCA fitted to this matrix retained $n = 3$ or 8 directions. To decode, inverse PCA restored 18 coefficients, then the spline basis evaluated nine x coefficients and nine y coefficients at the 100 phases. Validation reconstruction error selected raw or standardized coefficients. A separate Ridge regression predicted two durations from the latent scores and five task conditions. Conditions therefore affected spline timing but not its spatial decoder (Appendix A2).

Neural models received 200 coordinates standardized using training statistics. Encoder and decoder each used two 256-unit ReLU layers. VAE and CVAE encoders predicted latent means and variances, with sampling during training and posterior means for reconstruction. Their loss combined summed coordinate error, a timing-error term weighted by 20, and a KL penalty toward a standard-normal prior [1]. CVAE received the five task conditions at both encoder and decoder; VAE masked them to zero. CAE also received conditions, but used a deterministic code without the KL term. All decoded a trajectory and two durations. Timing targets used standardized $\log(t + 0.001)$ values and were never encoder inputs. Training used Adam, validation-based stopping and three seeds; complete equations and settings appear in Appendix A3.

3.2 Generation from personal context

Each test participant's trials were split approximately in half into context and query sets, stratified by starting category and side. The split was identical across models. We averaged context codes into a personal center, then drew 120 latent samples around it using a shared covariance estimated from training participants. The decoder transformed these samples into new paths and durations. This is a personal center with shared latent variability, not a separately fitted covariance for every participant (Appendix A4).

Generation withheld query trajectories and timing labels. Where a model used conditions, generation drew from the query-condition mixture for a matched-condition comparison. To test personal information, we replaced the center with the training-population center or other test-participant centers while holding the decoder, conditions and random draws fixed.

3.3 What each evaluation measures

Reconstruction: each decoded path was compared with its own observed input using mean squared coordinate error, $MSE = \text{sum of 200 squared coordinate differences} / 200$. This paired error checks information retained through compression. **Generation:** new samples have no uniquely matching recorded trial, so we compared distributions. Eleven features covered timing, speed, geometry, speed-peak count and endpoints. Mean KS averaged the largest empirical cumulative-distribution gap for each feature. Energy discrepancy [3] and Gaussian-kernel MMD squared [4] evaluated joint feature distributions after training-derived standardization.

The full-trajectory follow-up applied energy and MMD to the ordered 100-point paths, using RMS coordinate difference as the underlying distance and training-derived kernel bandwidth. An axis-balanced sensitivity scaled x and y by their training variation. Lower discrepancies indicate closer distributions; they are not accuracy percentages. Feature distances also depend on predicted durations, while complete-path distances assess normalized phase. Appendix A5-A6 gives the feature definitions and estimators.

Seeds were averaged within each participant before giving the 28 participants equal weight. Paired two-sided Pratt Wilcoxon tests used BH correction [5] over 140 original comparisons and a separate family of 80 full-path comparisons, with Holm sensitivity. These are exploratory comparisons with overlapping training rotations. A training-only feature-redundancy sensitivity and its separate correction family appear in Appendix A11.

4 Reconstructing an observed movement

Reconstruction asks whether an observed movement survives compression. Figure 2 overlays recorded and reconstructed paths for one trial, showing how the same input is represented at three and eight dimensions. Each model receives the complete observed trajectory for this task; successful reconstruction does not alone establish successful generation from a participant center.

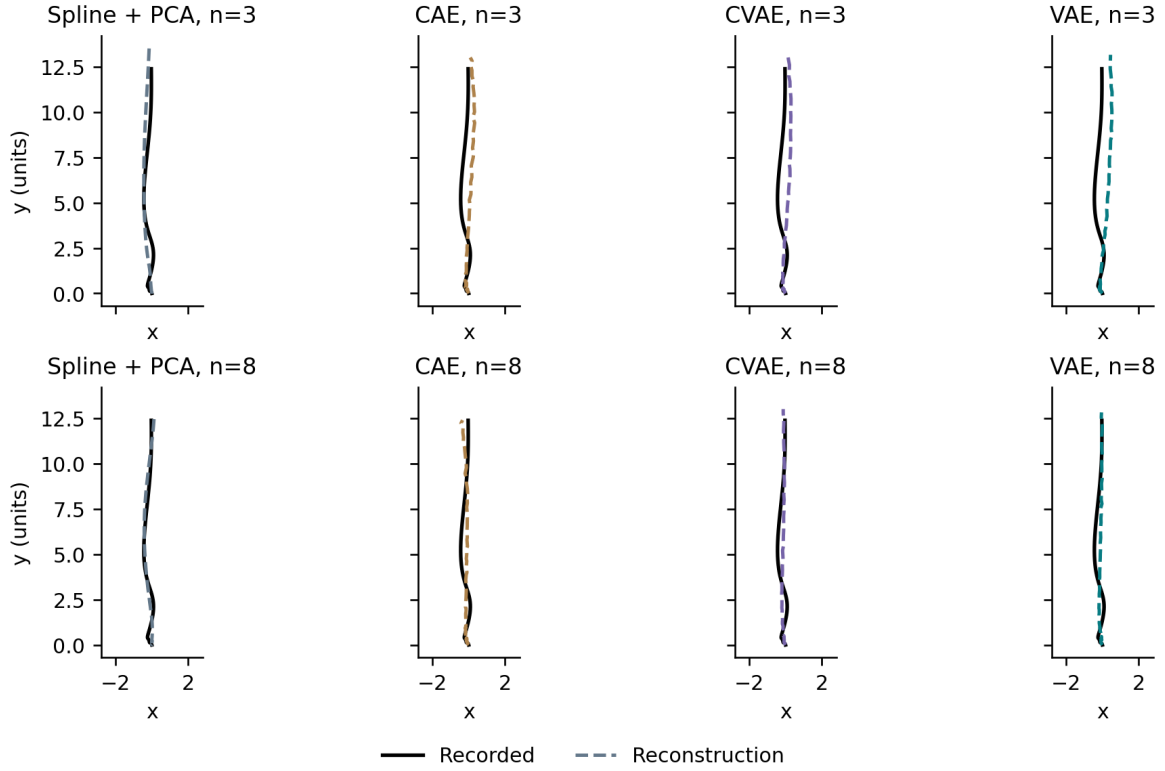


Figure 2. Recorded path (black) and reconstruction (colored dashed line) for subject01, selected by identifier and median modeled duration before inspecting model errors. Top: $n = 3$; bottom: $n = 8$. Each panel has equal x/y scale and shared limits. Neural examples use fold 2, seed 42. These overlays show spatial fidelity; coordinate-versus-phase curves in Appendix A12 also expose temporal progression. Statistical comparisons use all participants, not this example.

Table 1. Mean reconstruction MSE in squared tracker units; lower is better. Spline beat each neural model at each dimension under BH-140 (all $q < 0.0031$). Spline versus CAE $n = 8$ did not survive Holm ($p = 0.106$); the other five comparisons did. VAE led neural reconstruction at $n = 3$ under BH, while CAE led at $n = 8$. Condition-only Ridge MSE was 1.233795.

Model	$n = 3$ MSE	$n = 8$ MSE
Spline + PCA	0.097255	0.022772
CAE	0.169661	0.025968
CVAE	0.176981	0.032481
VAE	0.148299	0.033487

Spline + PCA had the lowest mean reconstruction error at both capacities. Eight-dimensional mean errors were smaller for every compact model, although no new between-capacity significance test was performed. PCA was therefore a strong compression baseline. Its coefficient representation retained observed shape accurately even though, as the following experiments show, sampling around averaged coefficients was less successful at reproducing the distribution of new movements.

5 Generating movement variability

5.1 Complete trajectory distributions

Generation tests whether a context-derived center supports new movements with appropriate variability. Figure 3 compares recorded query paths with unpaired generated samples. Comparing the full ordered paths tests details that may be missed by a selected list of kinematic features.

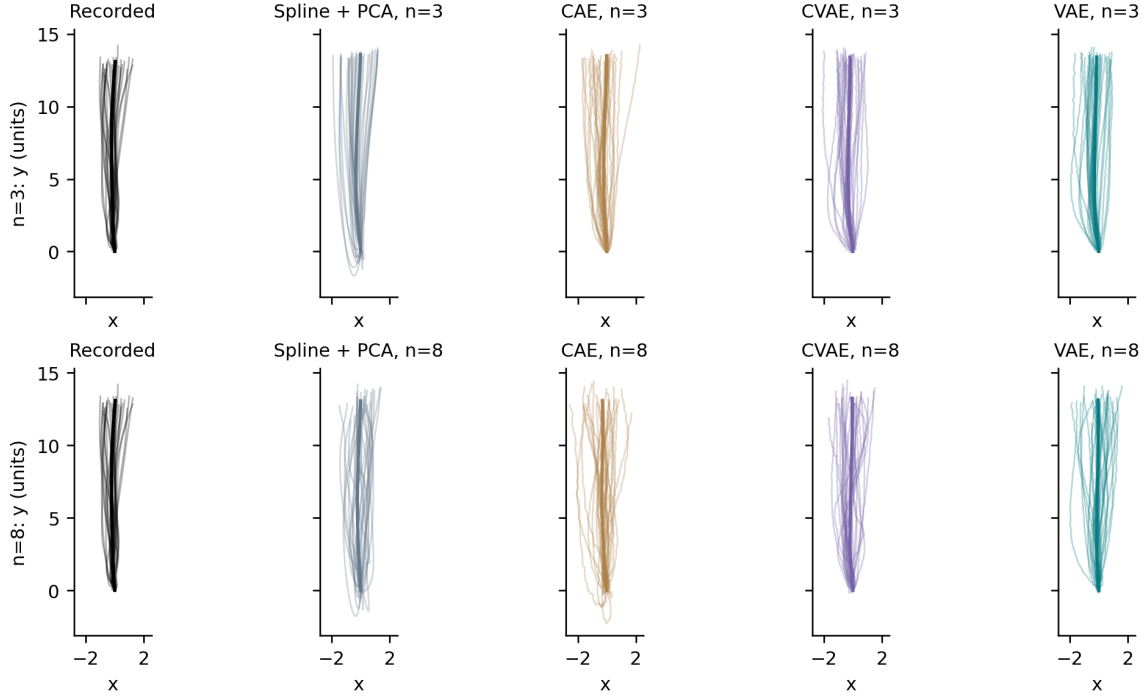


Figure 3. Subject01 query recordings and generated paths at $n = 3$ and $n = 8$, with equal x/y scale and shared limits. Thin curves show the first 30 stored paths; thick curves average all 90 query or 120 generated paths. Neural examples use fold 2, seed 42; participant selection did not use model scores. No generated curve has a uniquely matching recorded trial. Cohort inference is in Table 2.

Table 2. Complete-path discrepancies in RMS geometry; lower is better. At $n = 8$, VAE and CVAE each beat spline on both metrics under BH and Holm across 80 tests, also with balanced axes. VAE versus CVAE was not significant (BH $q = 0.672$ for energy, 1.000 for MMD). At $n = 3$, both beat spline on raw MMD ($q = 0.00184$; Holm $p = 0.0363$), but not energy, and those MMD advantages did not survive axis balancing. Full values appear in Appendix A8.

Model	n	Path energy	Path MMD squared
Spline + PCA	3	0.1068	0.0777
CAE	3	0.1272	0.0818
CVAE	3	0.1024	0.0578
VAE	3	0.1005	0.0572
Spline + PCA	8	0.0981	0.0705
CAE	8	0.0947	0.0652
CVAE	8	0.0549	0.0244
VAE	8	0.0548	0.0238

The strongest evidence favored the two variational models at eight dimensions. This differs from reconstruction because generation samples around a context average instead of encoding the particular query path. A representation can retain individual inputs accurately yet yield a less suitable distribution under its associated sampling procedure. The experiments establish this task-dependent ranking without identifying a single cause for it.

5.2 Kinematic-feature distributions

Feature comparisons ask whether generated paths preserve measurable aspects of the movements. The cumulative curves in Figure 4 show timing, speed and curvature directly. Joint distances complement these marginal views by comparing the complete feature vectors, including dependencies between features.

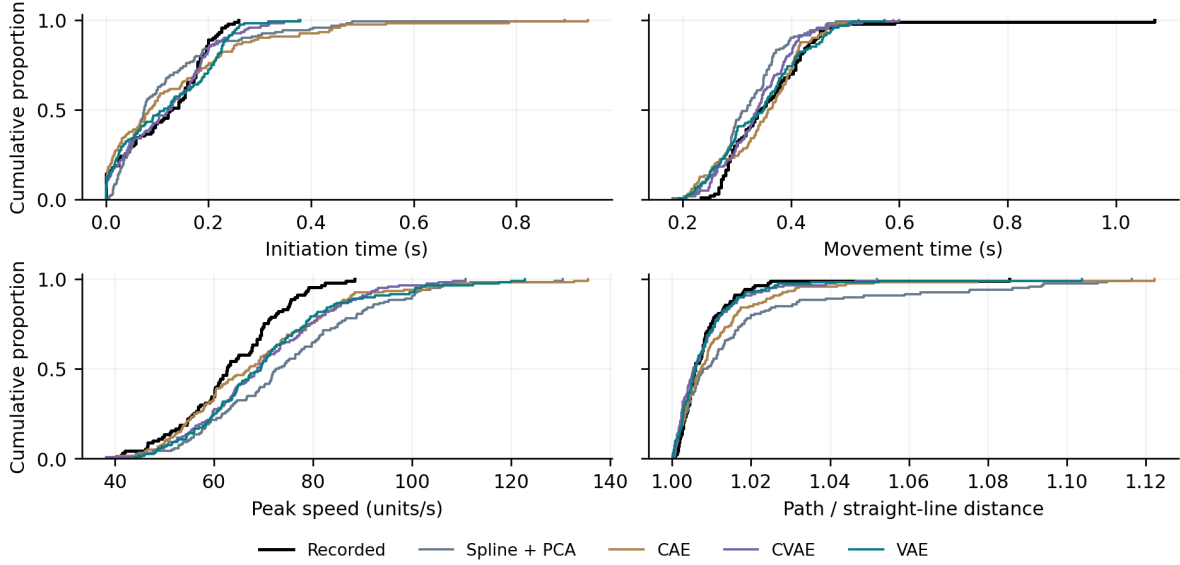


Figure 4. Recorded and generated empirical cumulative distributions for subject01 at $n = 8$ (fold 2, seed 42). A smaller gap from the black recorded curve indicates closer marginal fidelity. These four features were chosen for distinct meanings, not favorable scores. The cohort analysis includes all eleven features; Appendix A13 shows all curves at both capacities. Differences in this illustrative participant are not separate significance tests.

Table 3. Eleven-feature generation discrepancies; lower is better. VAE and CVAE beat spline on all three metrics at both capacities under BH and Holm-140. VAE $n = 8$ had the lowest means, but versus CVAE only energy passed BH ($q = 0.0257$), not Holm; KS ($q = 0.137$) and MMD ($q = 0.0518$) did not pass BH. Feature and path distances have different geometries and are not numerically interchangeable.

Model	n	Mean KS	Energy	MMD squared
Spline + PCA	3	0.3251	0.9774	0.1815
CAE	3	0.2876	0.8015	0.1414
CVAE	3	0.2778	0.7454	0.1309
VAE	3	0.2846	0.7476	0.1303
Spline + PCA	8	0.2548	0.7347	0.1153
CAE	8	0.2587	0.6202	0.1111
CVAE	8	0.2194	0.4305	0.0675
VAE	8	0.2151	0.4110	0.0639

A training-only redundancy screen identified straight-line distance and forward endpoint as strongly correlated with path length in every rotation. Removing these two features left nine. VAE and CVAE still beat spline on all three reduced-feature distances at both capacities under BH and Holm in the sensitivity family. However, VAE $n = 8$ no longer differed significantly from CVAE on any reduced-feature metric. Thus the advantage over spline was robust to this reduction, while a unique VAE advantage was not (Appendix A11).

The primary eleven-feature results remain reported to preserve the original evaluation. Correlation does not establish that a feature lacks scientific meaning. Moreover, our correction operates on model-comparison endpoints, not eleven independent KS p-values, so removing two features does not automatically remove two tests from the correction family.

6 Personal information, adequacy and the dashboard

A population generator could produce plausible movements without using information specific to the participant. Center replacement therefore tested whether the correct personal context improved the same generator. Own-context centers yielded better feature fidelity than either a population center or centers from other test participants (Figure 5).

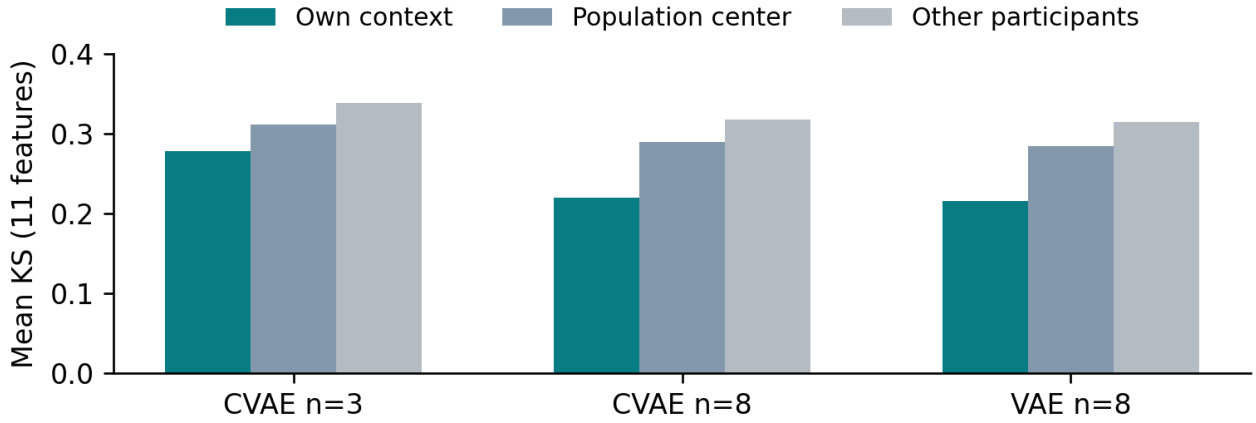


Figure 5. Mean KS after changing the latent center, with decoder, noise and task draws fixed. Lower is better. All six VAE $n = 8$ comparisons across KS, energy and MMD favored own context after BH correction ($q < 1.6 \times 10^{-6}$); all twelve CVAE $n = 3/8$ comparisons did so ($q < 0.000382$). This feature-based control supports personal information within a session; it was not repeated for the new full-path endpoints.

6.1 Relative improvement and absolute adequacy

Beating another model does not establish that a generator is sufficiently accurate for a particular application. An empirical reference that resampled complete stored context trajectories had lower mean full-path discrepancies than the compact models (Table 4). It retained more participant information than an n -value center, but showed that useful session information remained unrecovered by compression. No application-specific acceptance tolerance was established.

Table 4. Descriptive full-path reference means in RMS geometry. No significance test compared these empirical references with the models. Repeating a mean trajectory eliminates variability. Despite its better mean-path RMSE than VAE $n = 8$ (0.1146 versus 0.1604), its distribution discrepancies were much worse.

Generator / reference	Path energy	Path MMD squared
VAE $n = 8$	0.0548	0.0238
CVAE $n = 8$	0.0549	0.0244
Resample stored context	0.0328	0.0054
Repeat context mean	0.4621	0.2452

6.2 Interactive project deliverable

The dashboard integrates trajectory generation, model exploration and cohort benchmarks. It supports CVAE and spline + PCA at $n = 2, 3, 4$ and 8 , and CAE and VAE at $n = 3$ and 8 . VAE $n = 8$ is the default because it combines the lowest mean feature discrepancies with strong full-path results. CVAE remains a comparable generative alternative, while spline supplies the statistical benchmark and strongest mean reconstruction. This is a practical selection from the evaluated models, rather than a claim of uniform significant superiority.

The interface distinguishes illustrative outputs from cohort scores and exposes each model's conditioning behavior. VAE ignores explicit conditions; spline conditions affect timing only; CVAE and CAE receive conditions in both encoding and decoding. The dashboard enables inspection of model behavior, but its interactive controls do not independently validate a human response to changing a task condition.

7 Discussion

The results distinguish compression of an observed movement from generation of a movement distribution. Spline + PCA retained observed paths most accurately, whereas VAE and CVAE at eight dimensions produced closer complete-path and feature distributions under the tested generation procedure. Personal-center controls further showed that the correct participant's context improved that generator. The original three-coordinate objective was only partly achieved: compact personal context was useful, but the strongest and most robust generative evidence came from eight dimensions.

The feature-redundancy analysis refined this interpretation. Straight-line distance and forward endpoint overlapped strongly with path length, and excluding them preserved the variational models' advantages over spline. It did not preserve a significant VAE advantage over CVAE. We therefore retain VAE $n = 8$ as a practical dashboard default while treating CVAE as a comparable alternative. Feature reduction changes the evaluation geometry and should be justified by scientific coverage, rather than selected to obtain a smaller p-value.

The personal center was an average of context codes, with a shared latent covariance. It discarded information retained by direct context resampling. Direct context summaries also outperformed latent linear probes on all fourteen tested behavioral-summary targets (Appendix A10). Thus useful personal information for this decoder does not establish that a few coordinates fully represent every measured individual difference, nor that each coordinate has a stable physical interpretation.

Context and query came from the same session and a matched task-condition mixture. Cross-session stability and extrapolation to new conditions remain untested. Full-path distances describe normalized phase rather than physical time; feature distances depend additionally on predicted durations and the movement-segment definition. A good full-path result cannot establish accurate timing by itself. More broadly, the current comparisons provide relative evidence without a predefined threshold for application-level adequacy.

These analyses are exploratory. Participants were held out from fitting, but rotations shared training participants, three seeds do not exhaust training variability, and the dashboard default was selected using the evaluated cohort. The three-dimensional result depended on metric and axis scaling. Nonsignificant conditional-model advantages do not show that conditions are irrelevant. Independent validation after model selection and comparisons across sessions would strengthen the conclusions.

8 Conclusion

Compact context-derived representations support generation of participant-specific interception variability, with different models leading reconstruction and generation. VAE and CVAE at eight dimensions provide the strongest compact generation evidence relative to spline + PCA, including after reducing correlated features. The dashboard makes these comparisons accessible, while empirical resampling and unsuccessful summary probes identify information still lost through compression. Future work can investigate richer personal distributions and generalization beyond the observed session.

References

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